Configuração 6: M = 128, C = 8, K = 8, IoT = 20 dB, QPSK

Perda de SNR do BCD-MMSE (iter 1) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas

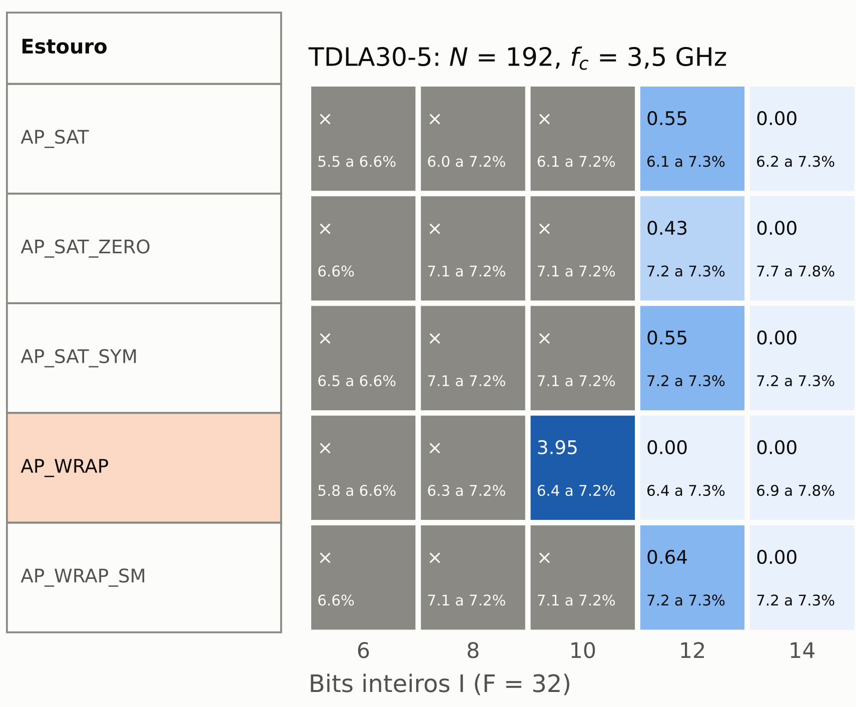
Fundo laranja no rótulo: o modo com o melhor resultado (estouro: pela varredura de I; quantização: pela varredura de F), comparando de menos bits para mais, sem contar a coluna em que tudo é zero

Faixa "x a y": o outro modo muda o valor (a quantização no mapa de I, o estouro no mapa de F); se os dois extremos têm cores diferentes, a célula é dividida na diagonal (em cima, a cor do menor; embaixo, a do maior)

Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo; sem número onde o formato e o modo não foram sintetizados

Perda de SNR (dB)

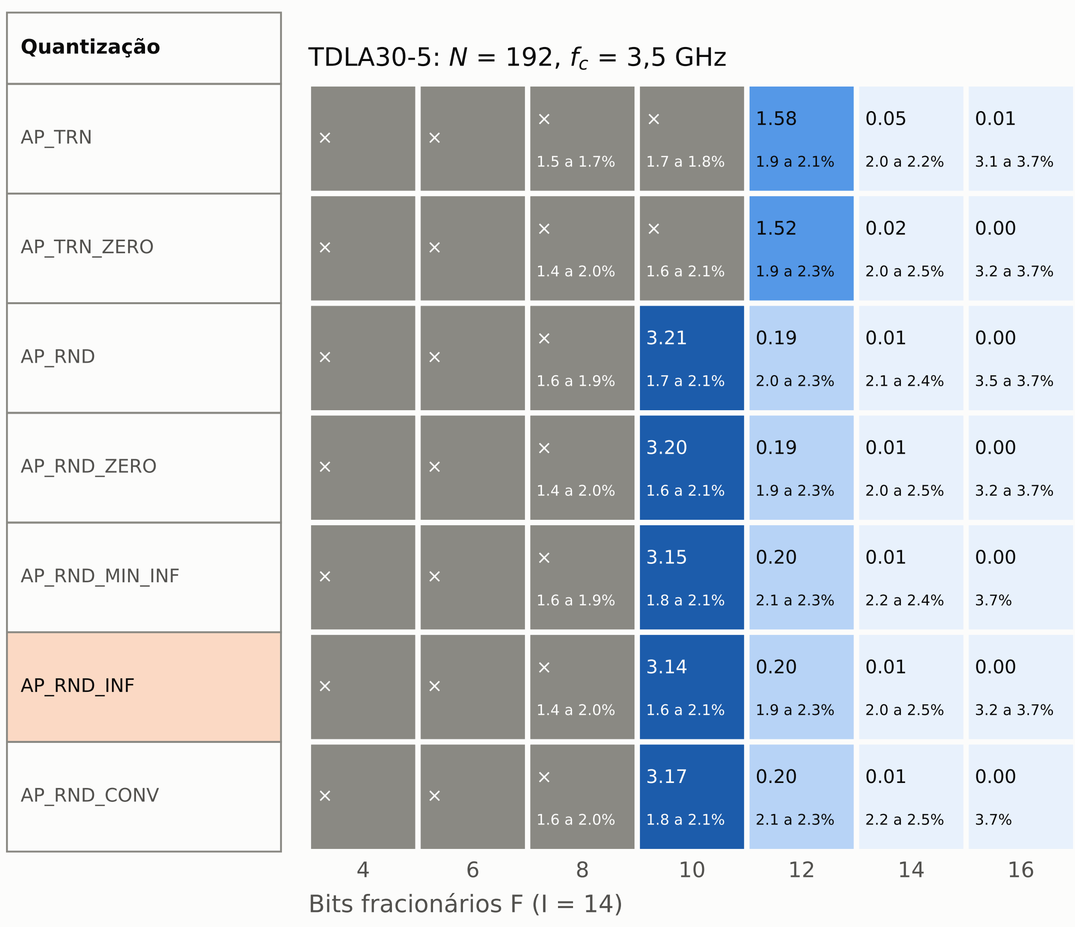
▭ < 0,1 dB ▭ 0,1 a 0,5 dB ▭ 0,5 a 1 dB ▭ 1 a 2 dB ▭ 2 a 3 dB ▭ 3 a 5 dB ▭ 5 a 10 dB ▭ > 10 dB ▭ × não chega a 10^{-2} ▭ rótulo do melhor modo



TDLA30-5: N = 192, f_c = 3,5 GHz

Estouro	6	8	10	12	14
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	× 6.1 a 7.2%	0.55 6.1 a 7.3%	0.00 6.2 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.43 7.2 a 7.3%	0.00 7.7 a 7.8%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.55 7.2 a 7.3%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	3.95 6.4 a 7.2%	0.00 6.4 a 7.3%	0.00 6.9 a 7.8%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.64 7.2 a 7.3%	0.00 7.2 a 7.3%

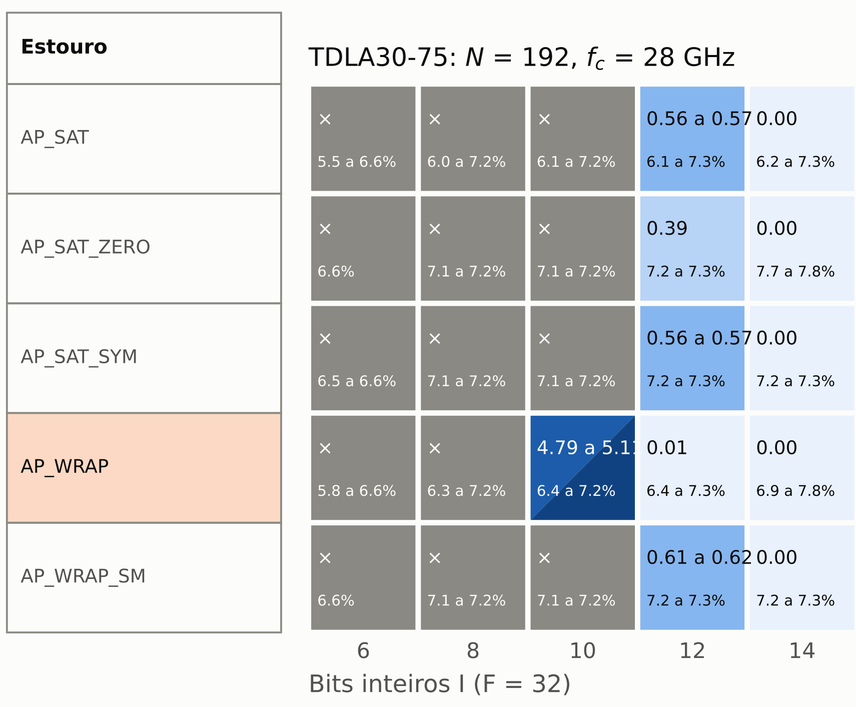
Bits inteiros I (F = 32)



TDLA30-5: N = 192, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.58 1.9 a 2.1%	0.05 2.0 a 2.2%	0.01 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.52 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	3.21 1.7 a 2.1%	0.19 2.0 a 2.3%	0.01 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	3.20 1.6 a 2.1%	0.19 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	3.15 1.8 a 2.1%	0.20 2.1 a 2.3%	0.01 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	3.14 1.6 a 2.1%	0.20 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	3.17 1.8 a 2.1%	0.20 2.1 a 2.3%	0.01 2.2 a 2.5%	0.00 3.7%

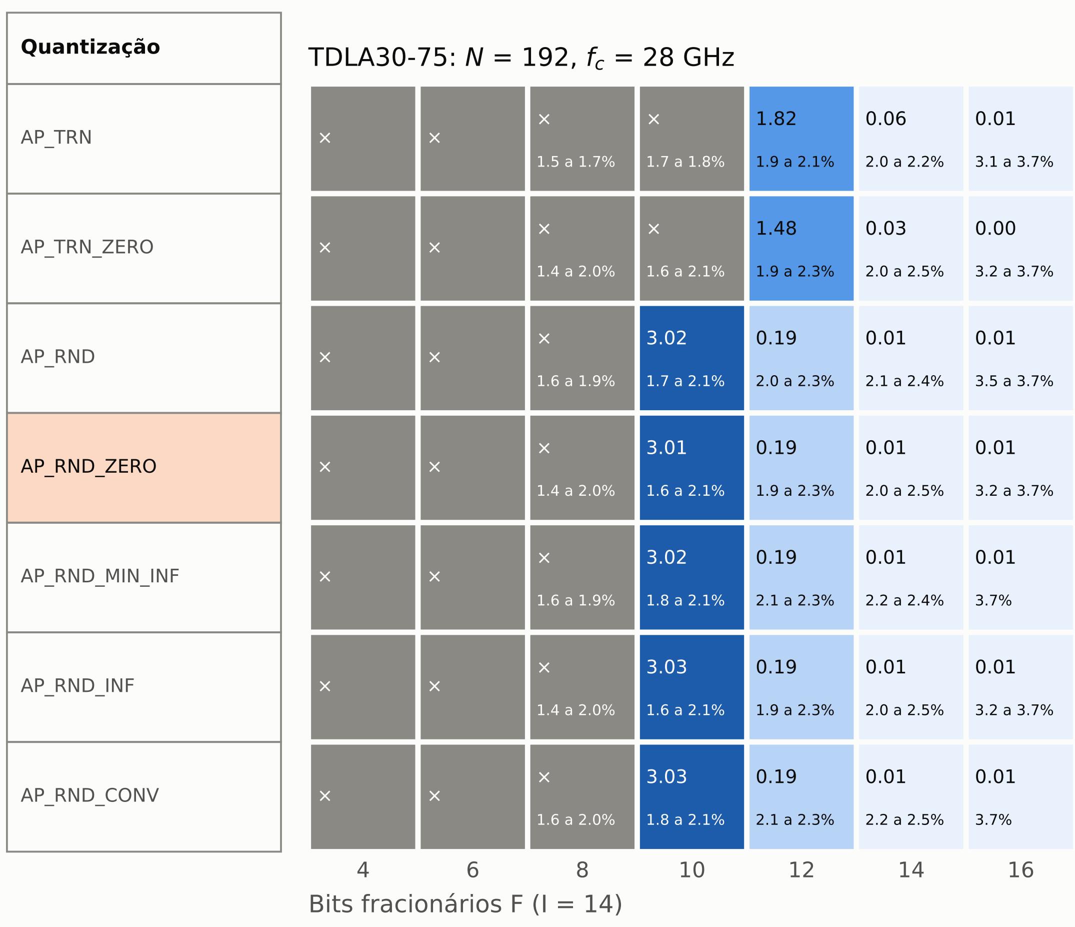
Bits fracionários F (I = 14)



TDLA30-75: N = 192, f_c = 28 GHz

Estouro	6	8	10	12	14
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	× 6.1 a 7.2%	0.56 a 0.57 6.1 a 7.3%	0.00 6.2 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.39 7.2 a 7.3%	0.00 7.7 a 7.8%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.56 a 0.57 7.2 a 7.3%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	4.79 a 5.1 6.4 a 7.2%	0.01 6.4 a 7.3%	0.00 6.9 a 7.8%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.61 a 0.62 7.2 a 7.3%	0.00 7.2 a 7.3%

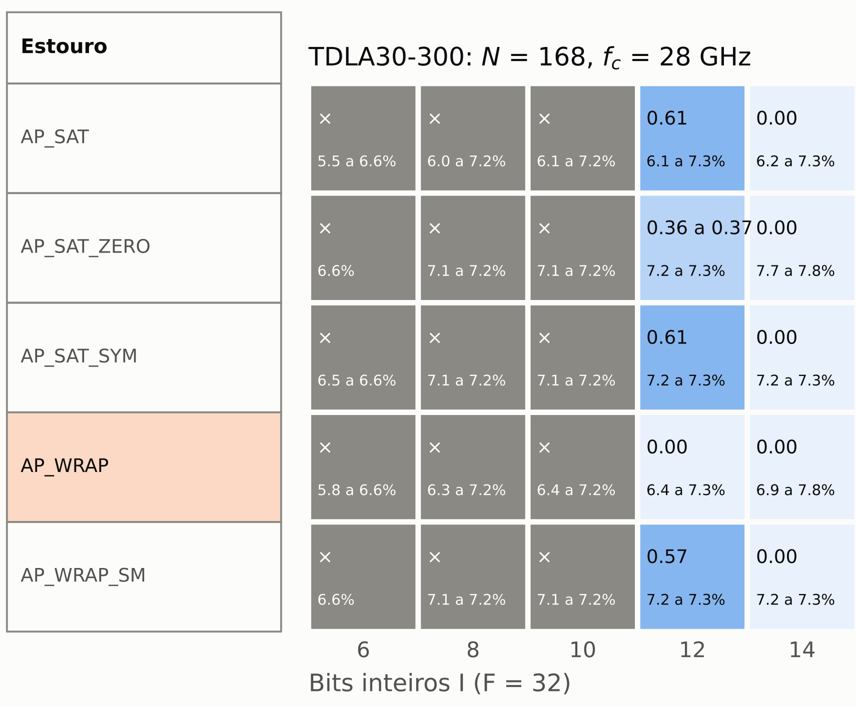
Bits inteiros I (F = 32)



TDLA30-75: N = 192, f_c = 28 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.82 1.9 a 2.1%	0.06 2.0 a 2.2%	0.01 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.48 1.9 a 2.3%	0.03 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	3.02 1.7 a 2.1%	0.19 2.0 a 2.3%	0.01 2.1 a 2.4%	0.01 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	3.01 1.6 a 2.1%	0.19 1.9 a 2.3%	0.01 2.0 a 2.5%	0.01 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	3.02 1.8 a 2.1%	0.19 2.1 a 2.3%	0.01 2.2 a 2.4%	0.01 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	3.03 1.6 a 2.1%	0.19 1.9 a 2.3%	0.01 2.0 a 2.5%	0.01 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	3.03 1.8 a 2.1%	0.19 2.1 a 2.3%	0.01 2.2 a 2.5%	0.01 3.7%

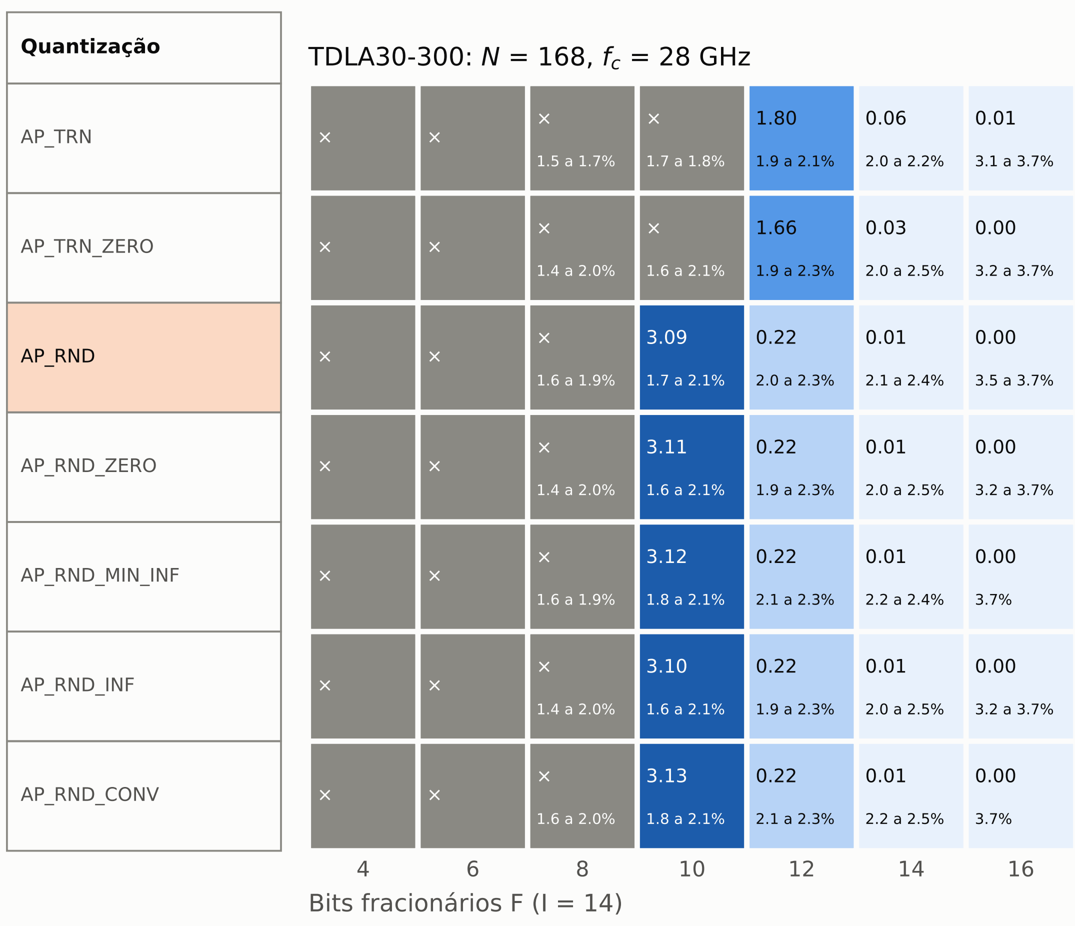
Bits fracionários F (I = 14)



TDLA30-300: N = 168, f_c = 28 GHz

Estouro	6	8	10	12	14
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	× 6.1 a 7.2%	0.61 6.1 a 7.3%	0.00 6.2 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.36 a 0.37 7.2 a 7.3%	0.00 7.7 a 7.8%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.61 7.2 a 7.3%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	× 6.4 a 7.2%	0.00 6.4 a 7.3%	0.00 6.9 a 7.8%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.57 7.2 a 7.3%	0.00 7.2 a 7.3%

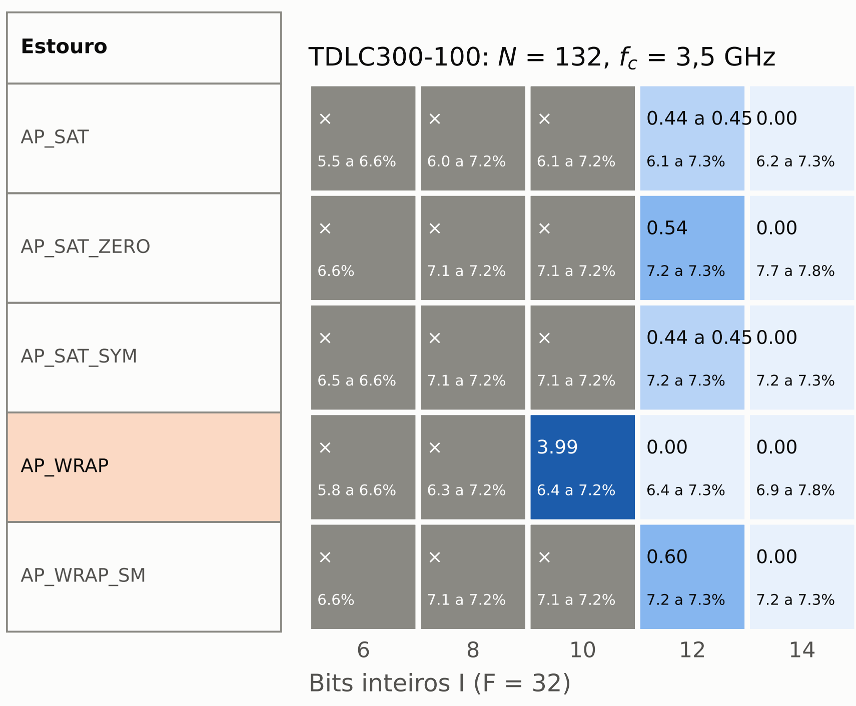
Bits inteiros I (F = 32)



TDLA30-300: N = 168, f_c = 28 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.80 1.9 a 2.1%	0.06 2.0 a 2.2%	0.01 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.66 1.9 a 2.3%	0.03 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	3.09 1.7 a 2.1%	0.22 2.0 a 2.3%	0.01 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	3.11 1.6 a 2.1%	0.22 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	3.12 1.8 a 2.1%	0.22 2.1 a 2.3%	0.01 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	3.10 1.6 a 2.1%	0.22 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	3.13 1.8 a 2.1%	0.22 2.1 a 2.3%	0.01 2.2 a 2.5%	0.00 3.7%

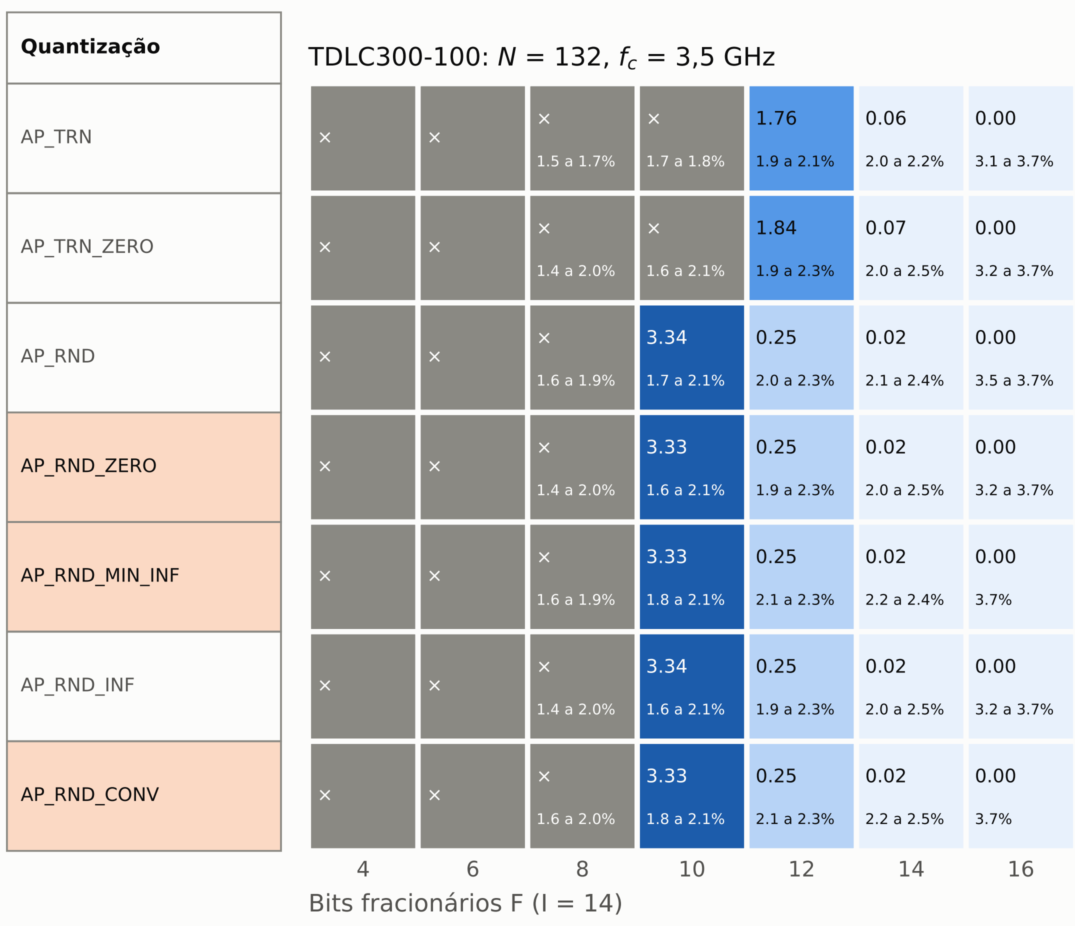
Bits fracionários F (I = 14)



TDLC300-100: N = 132, f_c = 3,5 GHz

Estouro	6	8	10	12	14
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	× 6.1 a 7.2%	0.44 a 0.45 6.1 a 7.3%	0.00 6.2 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.54 7.2 a 7.3%	0.00 7.7 a 7.8%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.44 a 0.45 7.2 a 7.3%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	3.99 6.4 a 7.2%	0.00 6.4 a 7.3%	0.00 6.9 a 7.8%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	× 7.1 a 7.2%	0.60 7.2 a 7.3%	0.00 7.2 a 7.3%

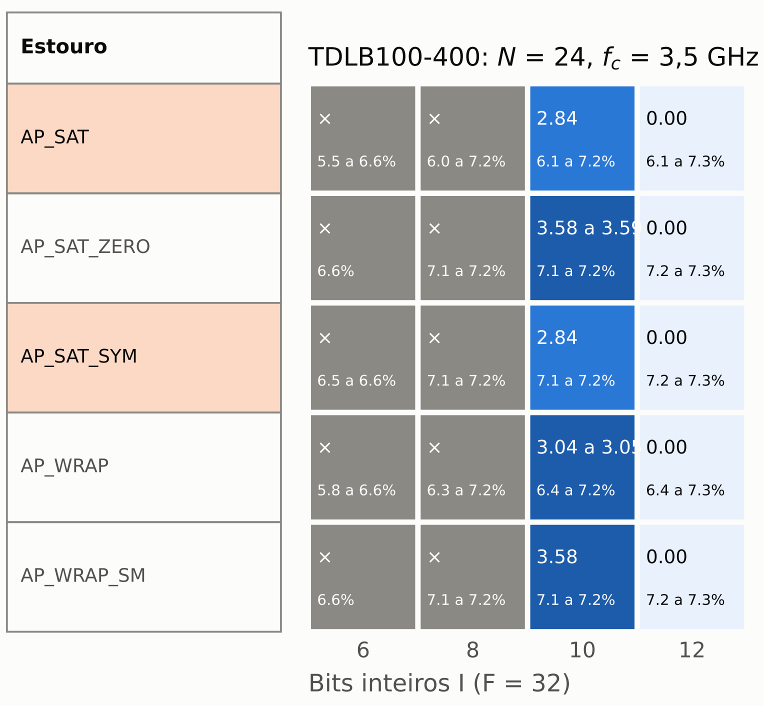
Bits inteiros I (F = 32)



TDLC300-100: N = 132, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.76 1.9 a 2.1%	0.06 2.0 a 2.2%	0.00 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.84 1.9 a 2.3%	0.07 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	3.34 1.7 a 2.1%	0.25 2.0 a 2.3%	0.02 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	3.33 1.6 a 2.1%	0.25 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	3.33 1.8 a 2.1%	0.25 2.1 a 2.3%	0.02 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	3.34 1.6 a 2.1%	0.25 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	3.33 1.8 a 2.1%	0.25 2.1 a 2.3%	0.02 2.2 a 2.5%	0.00 3.7%

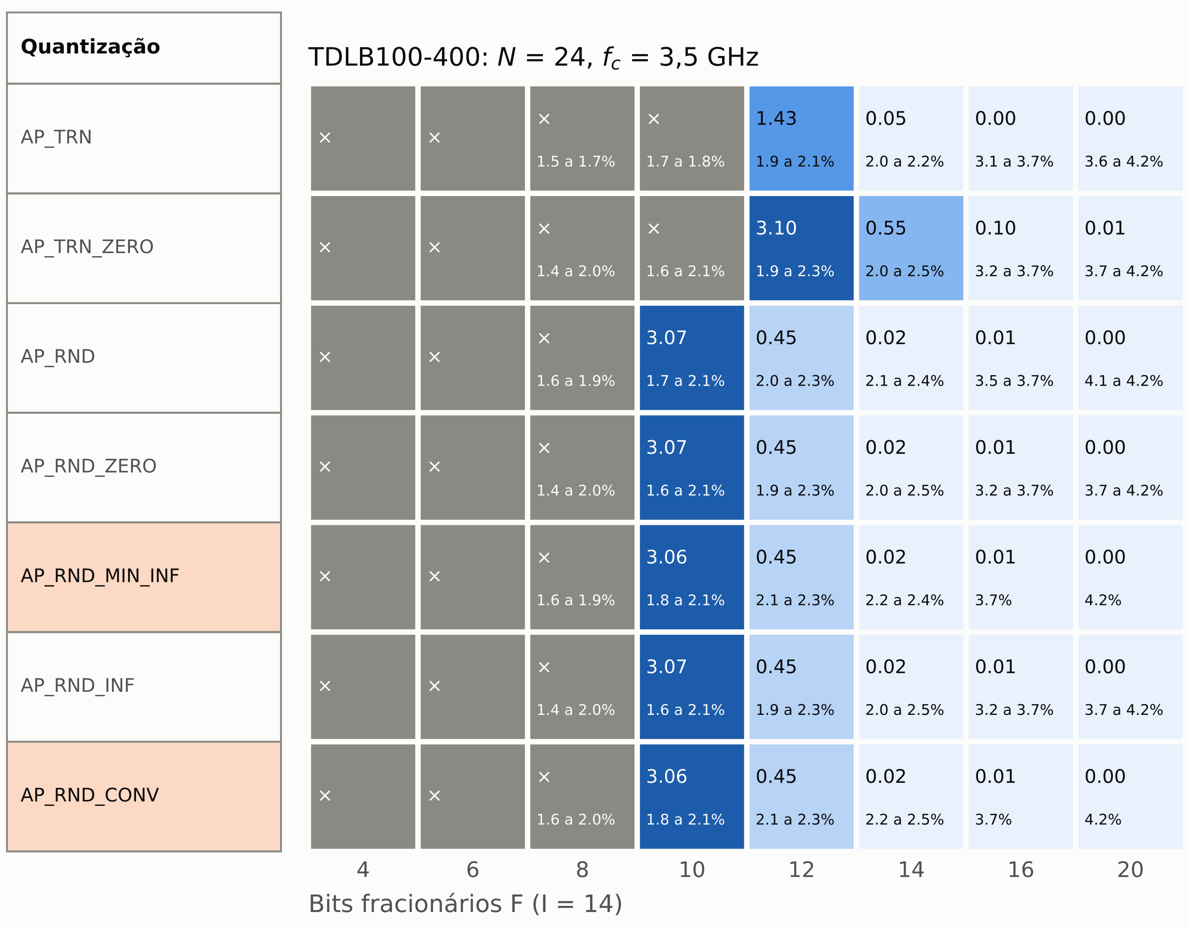
Bits fracionários F (I = 14)



TDLB100-400: N = 24, f_c = 3,5 GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.84 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	3.58 a 3.5 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.84 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	3.04 a 3.0 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	3.58 7.1 a 7.2%	0.00 7.2 a 7.3%

Bits inteiros I (F = 32)



TDLB100-400: N = 24, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14	16	20
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.43 1.9 a 2.1%	0.05 2.0 a 2.2%	0.00 3.1 a 3.7%	0.00 3.6 a 4.2%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	3.10 1.9 a 2.3%	0.55 2.0 a 2.5%	0.10 3.2 a 3.7%	0.01 3.7 a 4.2%
AP_RND	×	×	× 1.6 a 1.9%	3.07 1.7 a 2.1%	0.45 2.0 a 2.3%	0.02 2.1 a 2.4%	0.01 3.5 a 3.7%	0.00 4.1 a 4.2%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	3.07 1.6 a 2.1%	0.45 1.9 a 2.3%	0.02 2.0 a 2.5%	0.01 3.2 a 3.7%	0.00 3.7 a 4.2%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	3.06 1.8 a 2.1%	0.45 2.1 a 2.3%	0.02 2.2 a 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_INF	×	×	× 1.4 a 2.0%	3.07 1.6 a 2.1%	0.45 1.9 a 2.3%	0.02 2.0 a 2.5%	0.01 3.2 a 3.7%	0.00 3.7 a 4.2%
AP_RND_CONV	×	×	× 1.6 a 2.0%	3.06 1.8 a 2.1%	0.45 2.1 a 2.3%	0.02 2.2 a 2.5%	0.01 3.7%	0.00 4.2%

Bits fracionários F (I = 14)